

Deployment Models & LLM Strategy

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Deployment Models & LLM Strategy

SaaS · Private Cloud · Hybrid · On-prem — module-by-module, with the LLM/agentic story per model.
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The choice of deployment model is **not a feature toggle — it's a pricing tier and an architectural commitment**. SaaS is the default; everything else carries a price premium.

1. The four models — at a glance

Model	Where the data lives	Where the LLM lives	Where the agents run	Typical buyer	Price premium vs SaaS
SaaS (multi-tenant)	Hawkeye Vercel/Mongo Atlas	Cloud — Gemini Flash-Lite (free tier) + Claude Sonnet (paid)	Hawkeye serverless	SMB pharma, greenfield	baseline (1.0x)
Private Cloud (single-tenant)	Customer's AWS/Azure/GCP account	Cloud LLM via VPC endpoint	Customer's account	Mid-market with privacy/SOX	+50% to +100%
Hybrid	Customer's VPC for sensitive data; Hawkeye SaaS for non-sensitive	Cloud LLM with PII redaction proxy	Hawkeye SaaS for orchestration; sensitive call-outs go through customer's egress proxy	Pharma BigCo, EU GDPR sensitive	+30% to +75%
On-prem (airgapped or restricted egress)	Customer's data center	On-prem (Llama 3.1 70B / Mixtral 8x22B / Qwen 2.5 / Ollama)	Customer's data center	BigPharma · regulatory-restricted geos · airgapped CRO · government	+150% to +300%

2. SaaS (default) — *the only model we sell at SMB price*

Architecture today (live):

- Frontend: Next.js on Vercel (multi-tenant; tenant resolved per-domain or per-org-id JWT).

- Backend: Node/Express on Vercel serverless functions; one MongoDB Atlas cluster, sharded per tenant via `tenant_id` field; Postgres for marketplace catalog.
- Workers: Vercel cron for scheduled jobs (OVERDUE scan, EXPIRE scan, monthly scorecards).
- LLM calls: direct REST to Gemini API + Anthropic API.
- Auth: JWT-only today; OAuth + SAML on the deployment-models roadmap.

LLM strategy:

- **Default model:** Gemini 2.5 Flash-Lite (free tier) for high-volume, low-stakes agents (questionnaire pre-fill, observation drafter).
- **Premium model:** Claude Sonnet 4.6 for high-stakes drafting (audit report assembler).
- **Routing service:** `groundedGenerate.js` already routes per-feature with fallback.

Value prop:

- Zero infra setup. Buyer signs a credit-card → invite users → start using.
- AI agents always available, no extra ops.
- Price: **base SaaS price** (engine + 1 vertical pack from \$30k/yr).

When NOT to sell SaaS:

- Customer has a written policy "no production data outside our VPC" → upgrade to Private Cloud.
- Customer is a regulator-facing record-keeper for non-cloud-permitted geos (China · Russia · India CDSCO-stricted) → On-prem.
- Customer is a BigPharma with a procurement requirement for SOC 2 Type II + HIPAA BAA + EU SCCs + custom DPA → Hybrid (SaaS engine, customer-VPC data).

3. Private Cloud (single-tenant) — *for mid-market with stricter compliance*

What changes vs SaaS:

- One MongoDB Atlas cluster per tenant (or customer's own Atlas account).
- Vercel project per tenant (or customer's own AWS / Azure / GCP).
- LLM calls go through customer's VPC endpoint (Gemini via Google Cloud private connection, Claude via AWS Bedrock).
- Marketplace catalog (Postgres) can be either Hawkeye-shared (read-only sync) or per-tenant (no marketplace participation).

Module separation:

- **All EQMS modules** (Document Control · CAPA · etc.) — fully in customer's VPC.
- **Marketplace** — *optional* — if customer joins the marketplace, a unidirectional outbound sync writes anonymized supplier metadata + audit-report metadata to Hawkeye's shared Postgres. Customer can opt out and run a "private supplier directory" only.
- **AI agents** — same as SaaS, but routed through customer's VPC LLM endpoint. Agent code runs in customer's serverless project.
- **Public regulatory intel scrapers** (openFDA, EMA EudraGMDP, WHO PQ) — run in Hawkeye's shared infrastructure (no customer data) → push results into customer's VPC via API.

LLM strategy:

- Customer's choice: cloud LLM via VPC endpoint, OR on-prem LLM (rare for Private Cloud — usually customers in this tier are OK with cloud LLM through their VPC).
- **Token spending governance** — full Admin Panel (Doc 4) controls per-user agent quotas.

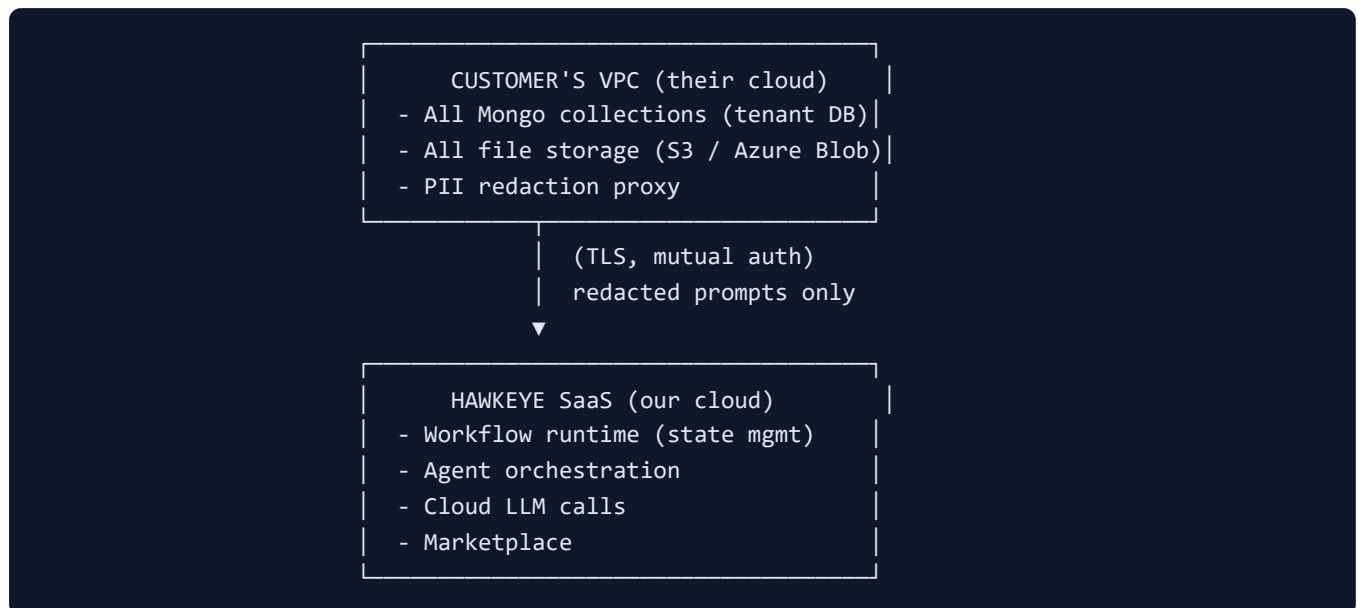
Value prop:

- Data residency (EU GDPR, India PDPB, etc.).
- SOC 2 Type II / HIPAA BAA / 21 CFR Part 11 audit boundary stays in customer's account.
- Customer's CISO can audit infrastructure access logs in their own SIEM.

Price premium: +50% to +100% over SaaS — covers per-tenant infra cost + ops overhead + SLA tier.

4. Hybrid — *for BigPharma with the "data stays in our VPC, AI can be cloud" stance*

The principle: split the architecture along the **data-vs-compute** seam.



Module separation:

- **Sensitive modules** (Batch Records · Deviation · Complaint with patient data · Manufacturing data) — data in customer VPC; only metadata + audit-trail summaries flow to Hawkeye.
- **Non-sensitive modules** (Supplier Pre-Qual · Audit · Document Control of public SOPs · Marketplace) — Hawkeye SaaS.
- **AI agents** — agent runtime in Hawkeye SaaS; **PII redaction proxy** in customer VPC scrubs PHI / proprietary process data before the prompt leaves the VPC.

LLM strategy:

- All LLM calls go to cloud LLM (Gemini / Claude).
- **PII Redaction Proxy** (the technical piece that makes this credible):
 - Runs as a Node service in customer's VPC.
 - Intercepts every prompt before egress.
 - Strips: patient names, lot numbers, supplier business secrets matching a customer-supplied dictionary, financial figures > threshold.

- Redacted prompt → cloud LLM → response → un-redacted in customer VPC if needed (lookup table in VPC).

- **Customer can audit every prompt** via the proxy log.

Value prop:

- BigPharma DPO can answer "no PHI ever leaves our VPC" while still using cloud LLM.
- Hawkeye gets to keep the marketplace + workflow OS as a managed service.

Price premium: +30% to +75% over SaaS — covers PII proxy infra + monitoring + DPA negotiation.

5. On-Prem — *the hardest sell, the highest price, the deepest moat*

This is the model that wins **BigPharma in restricted geographies, CROs with airgapped network policies,** and **government / defense-adjacent** opportunities.

What "on-prem" actually means

Three sub-flavors, each with different LLM implications:

Flavor	Egress	LLM lives where	Use case
Restricted egress	Allow-list of DNS / IPs (e.g., Hawkeye's API, customer's IDP)	Cloud LLM via allow-list	Mid-market customer with restrictive egress firewall
Customer-managed cloud	Customer's AWS GovCloud / Azure Gov / Alibaba Cloud (China)	Cloud LLM in same gov region (Bedrock GovCloud, Azure OpenAI Gov)	US gov · EU sovereign · China (Alibaba Qwen)
Airgapped (true on-prem)	No egress at all	On-prem LLM running on customer GPUs	Defense · DRDO/ISRO-adjacent · BigPharma R&D centers

The on-prem LLM stack — what we deploy

For airgapped on-prem, Hawkeye ships a **reference LLM serving stack** that the customer's IT can deploy:

Recommended models (one of):

1. **Llama 3.1 70B Instruct** (Meta) — best general-purpose, fits on 2× A100 80GB.
2. **Mixtral 8x22B Instruct** (Mistral) — strong reasoning, fits on 2× A100 80GB.
3. **Qwen 2.5 72B Instruct** (Alibaba) — best for China/Asia regulatory text.
4. **Gemma 2 27B** (Google) — smallest viable, fits on 1× A100 40GB or 1× L40S.

Recommended serving stack:

- **vLLM** for high-throughput batched inference (default for >10 concurrent users).
- **Ollama** for simple single-node deployments (≤5 concurrent users).
- **TensorRT-LLM + Triton** for NVIDIA-optimized installs (BigPharma data centers with H100s).

Hardware sizing reference (single tenant, 50 users, 1000 agent runs/day):

- Minimum: 1× NVIDIA A100 80GB → ~30 tokens/sec on 70B model, sufficient for batched workload with queuing.
- Recommended: 2× A100 80GB → ~80 tokens/sec, allows two agents concurrently.

- Premium: 1× H100 80GB → ~120 tokens/sec, allows real-time chat experience.

Serving infrastructure provided in the on-prem bundle:

- Containerized model serving (Docker / Kubernetes / Helm chart).
- LLM router that maps Hawkeye's existing `groundedGenerate.js` provider abstraction to the on-prem endpoint — zero code change in app code.
- Embedded vector store (Qdrant or Weaviate, on-prem) for RAG retrieval.
- **Vertical-pack-tuned prompt library** — Hawkeye ships pre-tuned prompts that perform well on Llama 70B (different from cloud-LLM prompts).

Agent feature parity on on-prem

Not all agents work equally well on Llama 70B vs Claude Sonnet. Honest delta:

Agent	Cloud (Sonnet/Gemini)	On-prem (Llama 70B)
Pre-Audit Questionnaire pre-fill	Excellent	Excellent (KB-grounded, low complexity)
Supplier-Intel public-data fusion	Excellent	Limited (requires public-internet fetches; needs proxied scraper or pre-indexed DB)
Audit Report Assembler	Excellent	Good (slightly less polished prose, regulatory citations may need second-pass review)
Observation Drafter	Excellent	Good
CAPA RCA Drafter (5-Whys)	Excellent	Excellent
Risk Brainstormer	Excellent	Good

The honest pitch to on-prem buyers: 5 of 6 agents perform at SaaS-grade quality on Llama 70B; Supplier-Intel requires either internet egress or a pre-indexed regulatory DB (Hawkeye ships a bundled openFDA mirror updated quarterly).

What's NOT in the on-prem bundle

Be explicit so the IT team trusts us:

- **✗ Marketplace** — networking effect requires SaaS multi-tenancy. On-prem customer can sync supplier-master data inbound from the cloud marketplace, but cannot publish into it.
- **✗ Public regulatory intel** — automated scrapers can't reach openFDA without egress. Hawkeye provides a quarterly-updated bundled DB.
- **✗ Auto-updates** — on-prem customer chooses when to install upgrades.
- **✗ Support telemetry** — opt-in only.

Pricing for on-prem

- **License fee:** \$80k-200k/yr per tenant (vs \$30-60k for SaaS).
- **Initial deployment fee:** \$50k-150k one-time (white-glove install + GPU procurement support + air-gap testing).
- **Annual support fee:** 20% of license/yr (M&S contract).
- **Optional:** per-GPU LLM serving licensing if customer wants Hawkeye to manage LLM ops.

6. Module-by-module deployment matrix

How each module behaves across the 4 models. Use this as the customer conversation.

Module	SaaS	Private Cloud	Hybrid	On-prem
Document Control	✓	✓	✓ in customer VPC	✓
Change Control	✓	✓	✓ in customer VPC	✓
Training	✓	✓	✓ in customer VPC	✓
Risk Register	✓	✓	✓ in customer VPC	✓
Internal Audit	✓	✓	✓ in customer VPC	✓
Management Review	✓	✓	✓ in customer VPC	✓
Deviation	✓	✓	✓ in customer VPC (PHI sensitive)	✓
CAPA	✓	✓	✓ in customer VPC	✓
Complaint	✓	✓	✓ in customer VPC (PHI sensitive)	✓
Supplier Pre-Qual	✓	✓	✓ Hawkeye SaaS (non-sensitive)	✓
Audit (8-phase)	✓	✓	✓ Hawkeye SaaS	✓
Equipment	✓	✓	✓ in customer VPC	✓
Batch Records	✓	✓	✓ in customer VPC (process IP)	✓
Marketplace	✓	✓ opt-in	✓ Hawkeye SaaS	✗ inbound-only
AI Agents	✓ Cloud LLM	✓ Cloud LLM via VPC	✓ via PII proxy	✓ on-prem LLM (5/6 parity)
Public Reg Intel	✓	✓	✓	⚠ quarterly bundled DB only
RFQ Procurement	✓	✓	✓ Hawkeye SaaS	✓ private RFQ only (no marketplace)

7. Sales playbook — which model to lead with by buyer profile

Buyer	Recommend
SMB pharma (50-500 emp)	SaaS . Period. Don't offer alternatives unless asked.
Mid-market pharma (500-2,000 emp) with EU operations	Private Cloud (EU region) — anchor at +50% premium.
BigPharma supplier-quality team (10-50 buyers)	Hybrid — sell as "your supplier-network layer, no PHI ever leaves your VPC."
BigPharma R&D / restricted geo (China, Russia, India CDSCO)	On-prem — engage their IT early; deployment is 8-16 weeks.
CRO / contract sterilizer / contract mfr	SaaS or Private Cloud depending on their customers' DPAs.
Government / defense-adjacent (rare)	On-prem airgapped , 6-12 month sales cycle.

8. What's next on the deployment-models roadmap

Quarter	Item
Q3 2026	Private Cloud deployment runbook + first paid Private Cloud customer
Q3 2026	Hybrid PII-redaction proxy (open-source the proxy, keep the prompts proprietary)
Q4 2026	On-prem reference architecture published (whitepaper + Helm chart)
Q1 2027	First on-prem paid pilot (target: BigPharma APAC R&D center)
Q1 2027	SOC 2 Type II audit complete (unblocks Private Cloud BigPharma deals)
Q2 2027	HIPAA BAA available + first hybrid med-device customer with PHI

9. The on-prem LLM honesty section

For your IT-team pitch (Doc 6 Track B), be ready to defend these claims honestly:

- **"Llama 3.1 70B matches Claude Sonnet for 5/6 of our agents."** Fair — based on internal eval. Can be reproduced with vendor evals (HumanEval, MMLU, MT-Bench).
- **"Customer's GPUs, customer's data, no egress."** True for the airgapped flavor.
- **"5/6 agent parity, NOT 6/6."** The honest delta — Supplier-Intel requires either egress OR a bundled openFDA mirror. Don't oversell.
- **"Setup is 4-8 weeks for a competent IT team."** True if they have GPUs procured. Add 8-12 weeks if they need to procure A100s.
- **"Hawkeye's app code is identical between SaaS and on-prem."** True — only the LLM provider URL changes via env var. This is the moat: the same workflow OS runs in both modes.

Don't claim:

- **"Same speed as SaaS."** False — on-prem 70B at ~30 tokens/sec vs Sonnet at ~80 tokens/sec.

- "All agents work identically." False — Supplier-Intel is degraded.
- "No GPU procurement needed." False — they need at least 1× A100 80GB or equivalent.